Lesson plan · for the teacher

Lists: a tool for your marks

Programming · Level 2 · Lesson 4

Time
40 min

Materials
Board and chalk; Ten made-up marks (never real pupils' marks); Your phone with this lesson downloaded

Learning goals

- Keep marks in a list and find them by position.
- Use len, sum, max and min.
- Loop over a list to count and average.

Activity: A human list

1. Warm-up (5 min): Five pupils stand in a row, each holding a made-up mark. Write marks = [...] on the board in the same order. Ask: how could we talk about one mark without a name?
2. Teach (10 min): Label the pupils 0 to 4 and explain that Python counts from 0. Write len, sum, max and min, and work out each one with the row.
3. Ask the list (10 min): Call out marks[2], marks[-1], max(marks). The pupil who holds the answer steps forward; the class checks.
4. Compare (10 min): Groups work out the average by hand while you run sum(marks) / len(marks) on your phone. Compare.
5. Wrap-up (5 min): Ask why counting from 0 causes bugs, and what marks[5] would do. Then give the exit ticket.

Use made-up marks only. Never put real pupils' names or marks on a shared board or a shared phone.

Discussion

- Why do you think lists count from 0?
- What would sorted(marks) tell you that max(marks) doesn't?
- Which list in our school would you like Python to summarise?

Exit ticket

- Write a list of three marks and the line that prints their average.

Try it in class tomorrow
Write ten made-up marks as a Python list on the board. Groups find the average and the top mark by hand while you run sum(marks) / len(marks) and max(marks) on your phone. Who finished first, and did you agree?

Worksheet · for pupils

Lists: a tool for your marks

Programming · Level 2 · Lesson 4

Name _____ Class _____ Date _____

1 **What does this program print?**

```
marks = [10, 12, 14]
print(sum(marks) / len(marks))
```

◯ 36 ◯ 12.0 ◯ 3

2 **The file marks.csv has a heading line, then one pupil per line, such as Amina,14. Finish the program: it asks for the pass mark and prints how many pupils reached it, like Passed: 3.**

```
pass_mark = int(input("Pass mark? "))
passed = 0
with open("marks.csv") as f:
    f.readline()  # skip the heading
    for line in f:
        name, mark = line.strip().split(",")
        # count the pupils who reached pass_mark
```

Write your program:

3 **Put the lines in order: the program prints every mark below 10, then the class average. One line doesn't belong.**

Number the lines in order (1, 2, 3…). Cross out any line that doesn't belong.

☐ `marks = [14, 9, 17, 6]`

☐ `for m in range(marks):`

☐ `for m in marks:`

☐ `print("Average:", sum(marks) / len(marks))`

☐ `if m < 10:`

☐ `print("Needs help:", m)`

4 **This program should print the last of the three marks, but it stops with an IndexError. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  marks = [14, 9, 17]
2  last = marks[3]
3  print("Last mark:", last)
```

---

5 **marks = [12, 15, 9]. Which of these give the class average, 12.0?**

- [ ] sum(marks) / len(marks)
- [ ] (12 + 15 + 9) / 3
- [ ] sum(marks) / 2
- [ ] max(marks) - min(marks)

✎ **Which list do you keep by hand (marks, attendance, book loans)? What would you ask Python to tell you about it: the average, the top mark, who needs help?**
